

World 0: no component acts

$$A = 0$$

$$B = 0$$



sum

$$Y = A + B + \varepsilon$$
$$V = h(|A + B|) + \eta$$

identical observables
(X, Y, V)

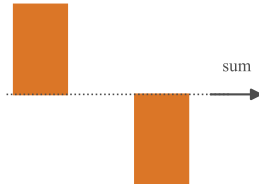
$$Z_{\text{component}} = 1$$

latent components

World 1: two components cancel

$$A = g(X)$$

$$B = -g(X)$$



sum

$$Y = A + B + \varepsilon$$
$$V = h(|A + B|) + \eta$$

identical observables
(X, Y, V)

$$Z_{\text{component}} = 0$$

latent components

Any rule reading only (X, Y, V) returns the same label in both worlds, yet component absence differs wherever $|g(X)| > \delta$.